ITAérea
Aeronautical Business School

Safety Management Systems for Airlines

Safety at Airlines

1. **Concepts of SMS**
2. **Aircraft Maintenance**
3. **IOSA - IATA Operational Safety Audit**
4. **Safety Culture**

SMS – Safety Management System



SMS – Safety Management System

1. **Concepts of SMS**
2. **What are the main 4 components?**
3. **Regulatory Framework**

1. Concepts of SMS

What is Operational Safety?

- **Operational safety is the state in which the risk of injury to persons or damage to property goods is reduced and maintained at an acceptable level,** or below, by means of a continuous process of identification of hazards and risk management



Refers to the measures and practices implemented to ensure safe and efficient flight op's. It includes various aspects, including:

- Flight Crew Training and Qualifications
- Aircraft Maintenance and Inspection
- Air Traffic Control (ATC)
- Emergency Procedures
- Risk Management and others

1. Concepts of SMS

Traditional approach on Operational Safety:

- **Compliance with regulations**
- **Oriented towards consequences**
- **Guilt or punishment for not fulfilling the duties of the safety**
- Accident prevention
- Unsafe acts by operational personnel
- Focuses on the security issues at hand



Identifies: **WHAT?** **WHO?** **WHEN?**

But it does not always reveal **WHY?** **HOW?**

1. Concepts of SMS

Prescriptive Regulation

Prescriptive Safety is based on compliance with the regulations/procedures in all areas of the transportation (technology, personnel, operation) and in the application of supervisory mechanisms (inspections and audits) to ensure its fulfillment

Prescriptive Safety is based on these principles:

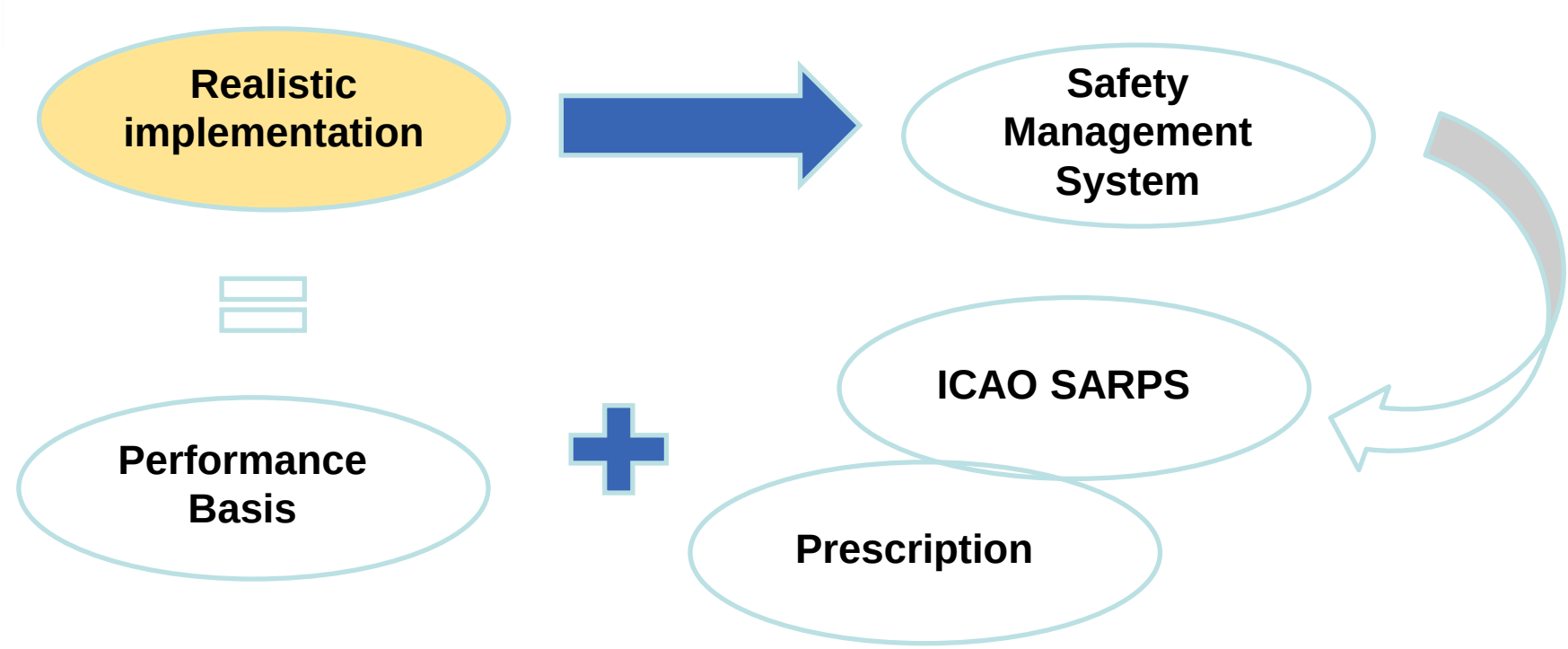
- **Transportation operations can be fully anticipated** during the planning, including all possible eventualities and interactions that may occur in the operational scenario.
- Technology, procedures and instruction can be provided **to address all operational eventualities.**
- Once operations have started**, safety is guaranteed if personnel operation meets the requirements** as established

Challenges with Prescriptive Regulation:

- **Barriers to Innovation** as strict adherence to prescriptive rules can hinder the adoption of new technologies.
- **How to incorporate non-Traditional Participants**
- **Emerging new aviation technologies require flexible regulatory approaches**

1. Concepts of SMS

There may be a set of circumstances where actual performance deviates from the designed performance, as a result of factors not under the direct control of the organization.

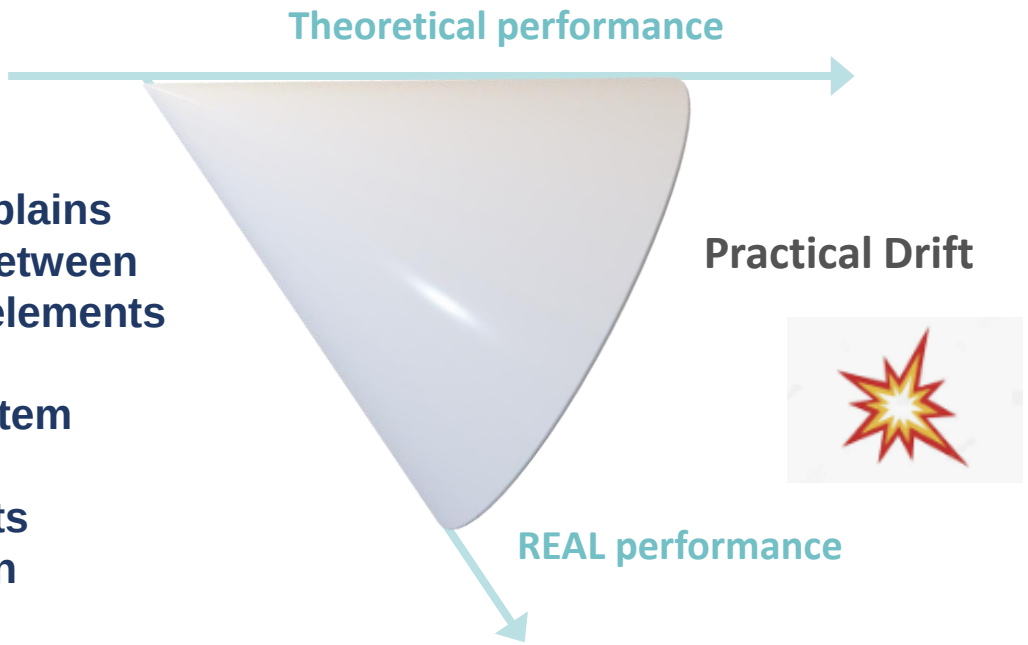


SARPS Standards And Recommended Practices

1. Concepts of SMS

Practical Drift

- **Practical Drift explains the decoupling between the prescriptive elements of the safety management system** (regulations and procedures) **and its implementation in practice**



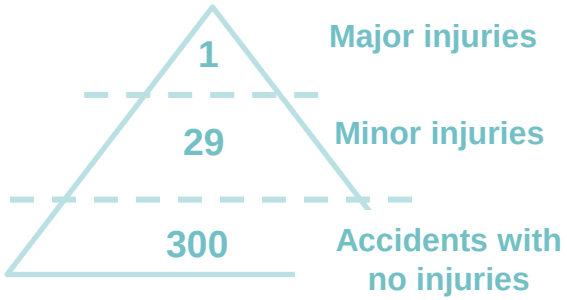
- There are many reasons that explain this phenomenon; technology that does not operate as intended, procedures not adapted to real environments, personnel violations, etc.

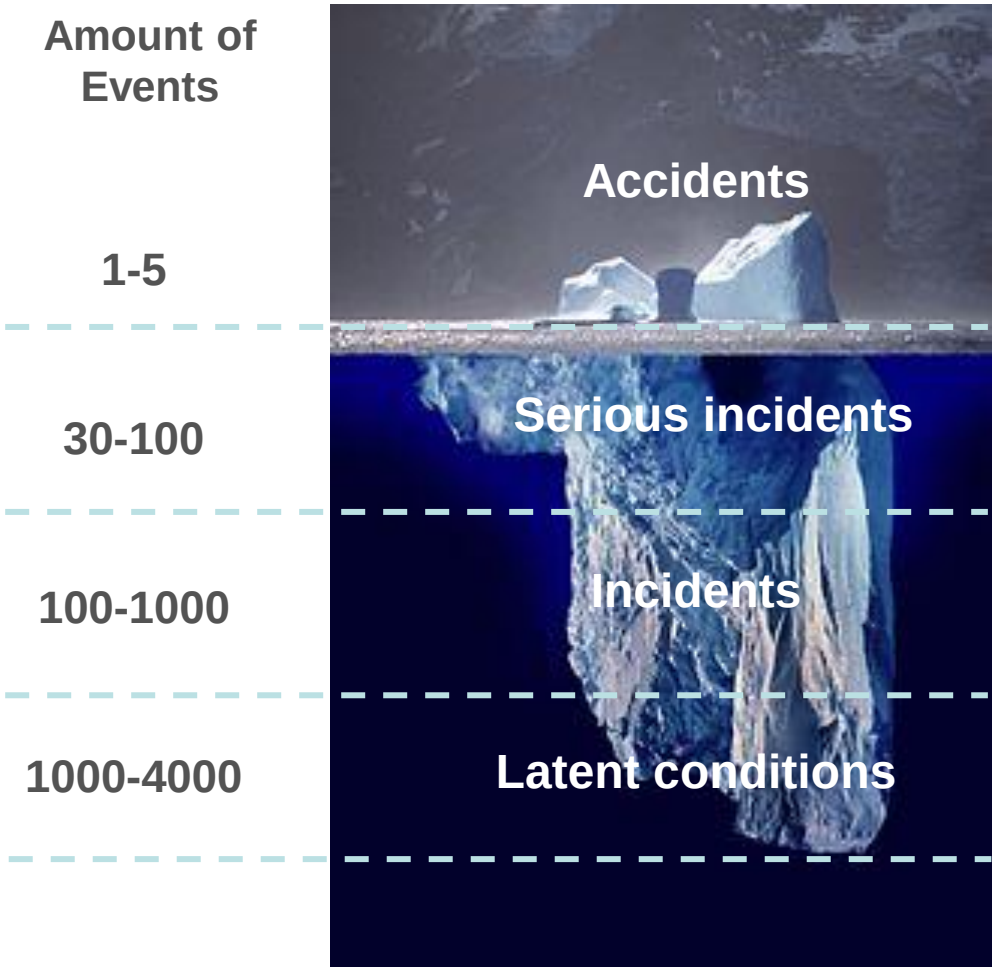
2. Concepts of SMS

There are usually a much bigger number of minor incidents happening before a major accident takes place.

The Heinrich´s triangle suggests a relationship between the number of minor and serious accidents.

Although the ratios may be discussed (1-29-300), it´s a key concept generally accepted





1. Concepts of SMS

Strategies

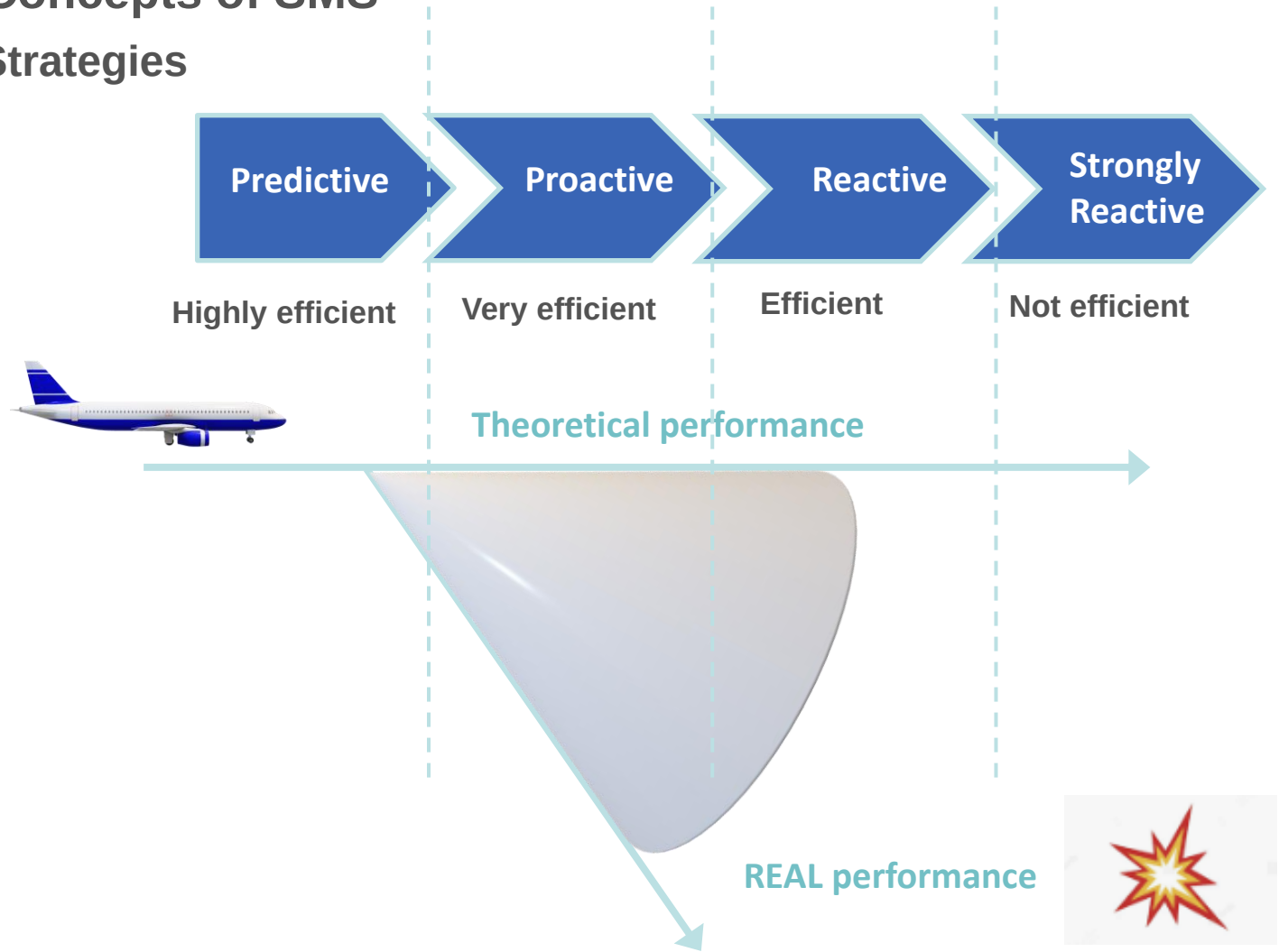
Depending on the time horizon, **strategies can be Reactive, Proactive or Predictive**

Reactive	Proactive	Predictive
Response to events that occurred (incidents and accidents)	Active search for dangers through analysis of the activities of the organization	Capture staff performance and what really happens during daily operations
“Wait for something to break”	“Fix problems without consequences"	“look for problems without waiting for them to occur"



1. Concepts of SMS

Strategies



1. Concepts of SMS

System	**Organized set of principles, resources, processes and procedures**
Management	**Resource Allocation / Prioritization**
Safety	**Condition by which risks are reduced to acceptable levels**

"Systematic, explicit and detailed process for managing security risks" (IATA, 2001)

"Systematic approach to security management that includes organizational structures, responsibilities, policies and procedures" (ICAO, Annex 19, July 2016)

1. Concepts of SMS

What is a Management System?

- The ISO standard defines it as a set of interrelated elements that establish policies and objectives, as well as the way to achieve them.
- It can also be defined as the **structure**, **processes** and **resources** necessary to establish the policy and objectives of an organization, as well as the way of achieve those goals



The SMS presents specificities that make it unique, and need to be overlapped with other management systems

1. Concepts of SMS

Organization size

The SMS must be proportional not only to the size, but also to the complexity of the organization and its activities.

In its dimensioning it should be considered:

- Organizational structure and resource availability
- Size and complexity of the organization
- **Complexity of activities and interfaces with third parties**



Complexity can come from:

- *Type of operation (off-shore, SAR, combat fire protection, etc.)*
- *Environment (mountainous terrain, extreme climate, fogs, airspace, etc.)*
- *Mixed fleets & Operational bases,*
- *Number of licenses, qualification, technologies used, number of subcontractors and outsourced tasks*

***For a small organization**, SMS appears as something complicated and/or expensive. However, SMS is simpler than it seems. It is about:*

- ***Actively looking for dangers in operations, products and services***
- *Applying corrective actions to control those dangers*
- *Monitoring the effectiveness of corrective actions implemented*

2. What are the main 4 components?

4 main components

There are four main components that will help us achieve a successful implementation of SMS.

1. Safety policies and objectives.
2. Safety risk management.
3. Ensuring operational safety.
4. Promotion of operational safety.



2. What are the main 4 components?



2. What are the main 4 components?



1

Policies & Objectives

The first component to the implementation of an SMS is **the organization's safety policies and objectives.**

The airline must create an environment where safety management can be effective, based on a safety policy and objectives set out by senior management's commitment to safety

Responsible executives and middle managers have, among others, the following obligations:

- Ensure that appropriate measures are taken to address safety issues and risks
- Respond to accidents and incidents.
- Allocate human, technical, financial or other resources for the effective and efficient operation of the MSS.
- **That personal safety responsibilities and roles are documented** and communicated throughout the org.

2. What are the main 4 components?



2

Risk Management

The second component to the implementation of a successful SMS is **risk management.**

Safety **Risk Management** is a fundamental component and **includes hazard identification and the assessment and mitigation of safety risks**

Hazard identification is the first step in the risk management process, precedes the assessment of safety risks, and requires a clear understanding of hazards and their related consequences.

The risk management process systematically identifies hazards that exist in the context of the delivery of our services, and **hazards may be the result of poorly designed systems. technical function, human interface, or interactions with other system processes**.

2. What are the main 4 components?



3

Ensuring Operational Safety

The third component of the GMDS **is safety assurance.**

Assurance involves establishing a systematic process for observing and measuring the organization's safety performance.

This includes activities such as **accident investigation**, change management, monitoring analysis and continuous improvement.

How and why accidents happen. In theory, it is established that accidents have multiple factors and causes, involving many people from different levels of the organization

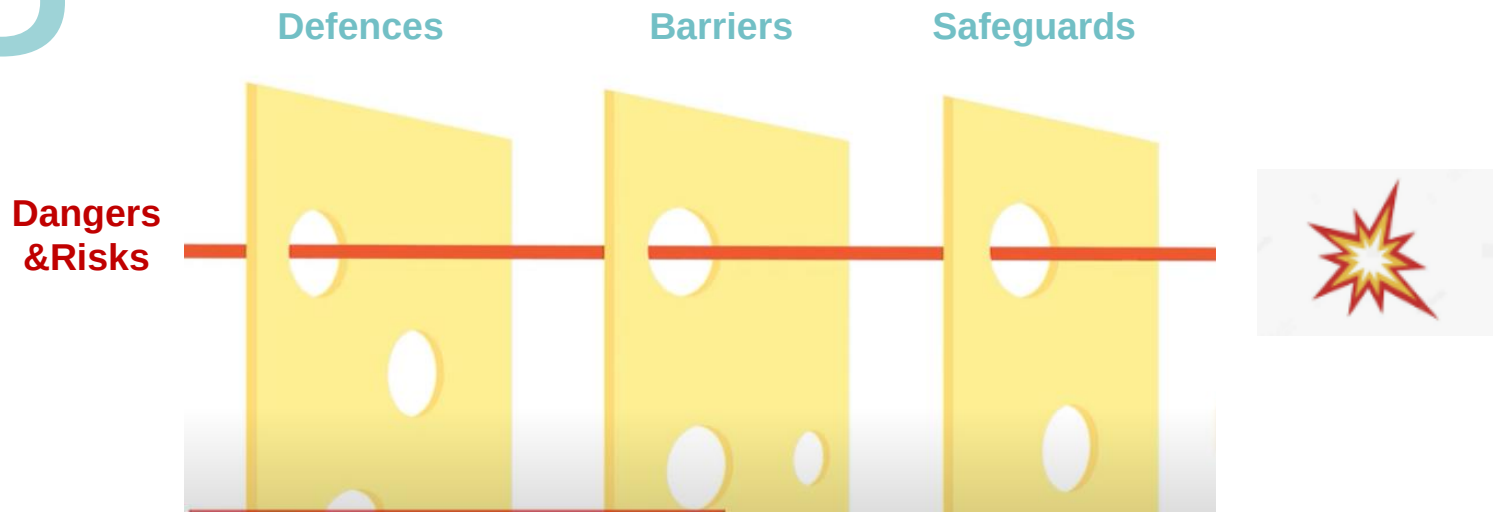
- The first is the organizational factor
- The second critical factor is working conditions.
- The third factor corresponds to unsafe acts that occur when there are errors or violations committed by staff during their working hours

2. What are the main 4 components?

3

Typically, the failure of one of these critical factors does not result in an accident.

However, the failure of all of these systems at the same time can create the opportunity for an accident, creating what is known as the Swiss cheese model.



Accidents mostly show that one person is usually not the sole cause of an accident. It's usually the result of a combination of mistakes. While we can't control other people's actions, there are many actions each of us can take to prevent the gaps in Swiss cheese from aligning.

2. What are the main 4 components?

Promotion of Operational Safety

4

The fourth component of MSS is the **promotion of safety.**

Effective advocacy and trainings like this foster awareness and understanding of SMS throughout the organization and help create a healthy safety culture.



It sets the tone for the behavior of each of the people in the organization.

Safety training provides skills and knowledge, as well as increases awareness of risk issues.

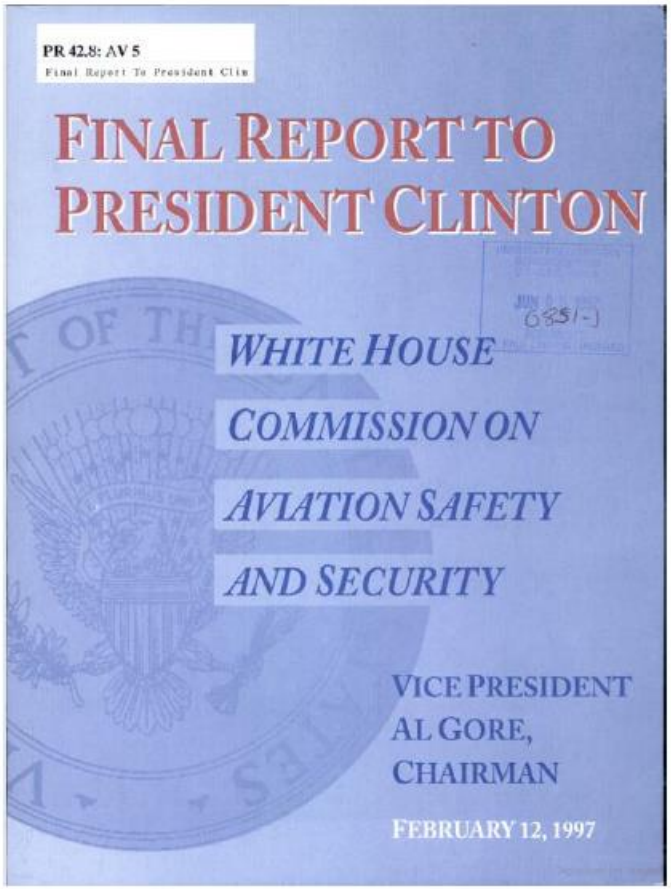
Training help the organization equip the culture that goes beyond simply preventing accidents or reducing the number of incidents.

Rather, it helps to construct a culture of doing the right thing at the right time, in response to both normal and emergency situations

3. Regulatory Framework

Background

- **In August 1996, US President Clinton established a Commission** made up of 25 members of the sector public and private, with the goal of preparing a safety strategy with National and International scope.
- The Commission completes its tasks in February 1997 and proposes as a global operational safety objective reducing the number of accidents by a factor of 5 in the following decade.
- To get to that goal it focuses:
  - On restructuring the regulatory framework and FAA certification programs to achieve that goal
  - **On improving collaboration with the industry in such a way that the Administration sets the objectives and works with the industry** on the best way to achieve them.



3. Regulatory Framework

ICAO: Assembly Resolutions

The White House Commission's recommendations approach was incorporated into ICAO in 1998. In 2001, the ICAO assembly Organization approves a resolution that includes the establishment of a framework for Safety Management

Resolution A32-15 of the 32nd Assembly from ICAO (1998)

- Improve databases and exchange of safety info.
- Promote safety communication among aviation users, including reporting accidents and incidents
- Urge States to ensure that the service providers have reporting volunteer systems

Resolution A33-16 of the 33rd Assembly (2001)

- Develop a aviation safety civil management framework, including recommendations for the improvement of safety and providing assistance



3. Regulatory Framework

Overall view

Global



ICAO Annex 19

US

FAA

European Union

EASA (AMCs)

National

National Legal Framework

** AMCs Acceptable Means of Compliance are non-binding standards adopted by EASA to illustrate means to establish compliance with the Basic Regulation and its Implementing Rules.*

3. Regulatory Framework

ICAO's strategic safety objectives are developed also through global, regional and national action plans (GASP, RASP and NASP)





Global Plan (GASP)

- Strengthen the State's safety supervision capacity
- Ensure that all States have implemented the SSP
- Increase collaboration at the regional level

Regional Plans (RASP)

- Strategic guide for safety management at the regional level
- Coordination of all regional initiatives aimed at improving safety
- Risk reduction and improved safety oversight/compliance
- Effective application of SMS and SSP by all States

National Plans (NASP)

- Define a strategic direction for safety management at the national level
- Establish the framework for safety management by the State
- Determine the State's action plan

National Legal Framework

3. Regulatory Framework

Other standards

- **The associations of aerospace industries of Europe, US, Brazil and Canada have developed in April 2022 a standard for the implementation of SMS in design, manufacturing and maintenance suppliers**.
- The standard is consistent with the regulations issued by national authorities (EASA, FAA, etc.) and ICAO, among others
- The structure of SMS requirements is fully aligned with that established by ICAO in Annex 19, and is accompanied by a guide that facilitates its interpretation and means of compliance
- A set of Annexes and Appendices completes the standard



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International Industry Standard

IMPLEMENTING A SAFETY MANAGEMENT SYSTEM
IN
DESIGN, MANUFACTURING AND MAINTENANCE ORGANIZATIONS

Publishers:

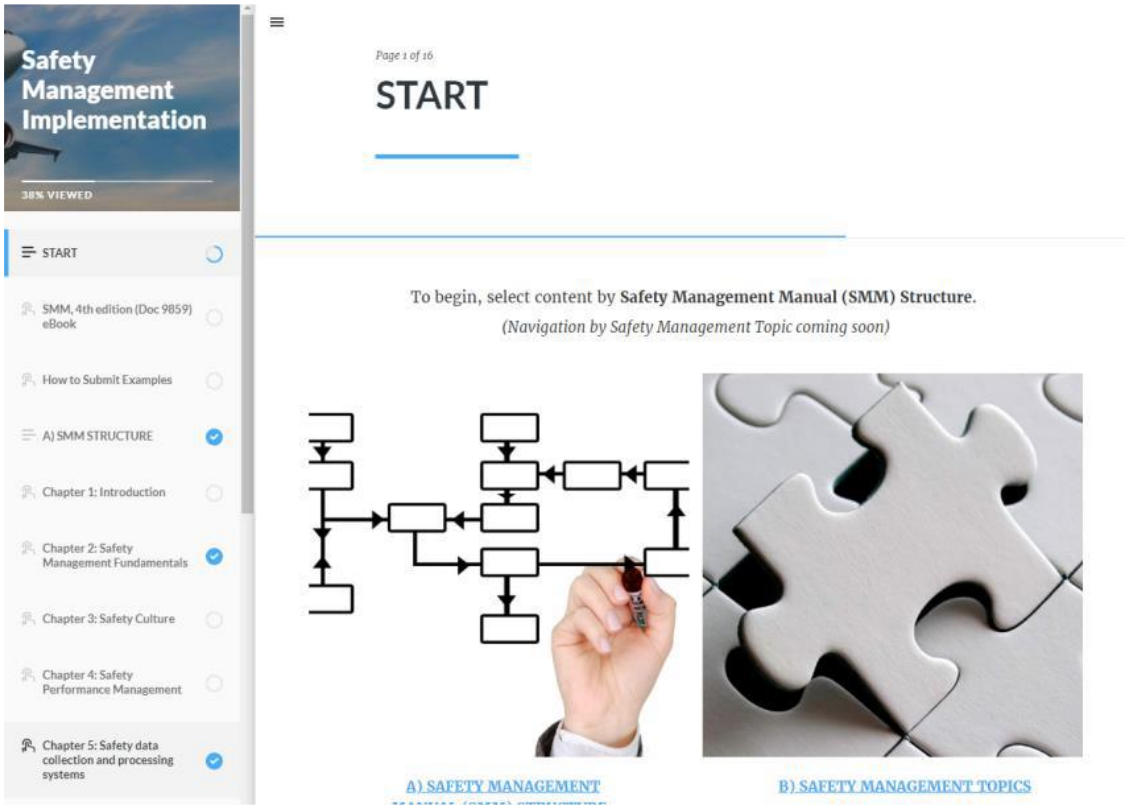
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3. Regulatory Framework

ICAO: Guidance material



ICAO SM Website

• ICAO has developed a website called **Safety Management Implementation** that complements SMM (Safety Management Manual) and provides examples, applications and supporting educational material to address the diverse needs of aviation community

• With this initiative, ICAO hopes to share the best practices with all those involved in the SMS/SSP implementation.

3. Regulatory Framework

EASA: Annual Safety Review (ASR)

- **The Annual Safety Review (ASR) is a comprehensive report that examines the safety performance of the European aviation system**
- It is prepared by the European Union Aviation Safety Agency (EASA) in collaboration with National Aviation Authorities (NAAs) and industry stakeholders
- The ASR identifies risks and, where necessary, includes mitigation actions in the EPAS



EASA: European Aviation Safety Plan (EPAS)

- **The EPAS provides a coherent framework for the improving security at EU level, supporting to the Member States in the application of their State Safety Programs (SSP)**
- The EPAS has a scope of four years and is updated annually
- The plan is structured in three volumes:

  Volume I: Strategy & Volume II: Actions & Volume III: Risk Portfolios Safety



3. Regulatory Framework

EASA/UE : Acceptable Means of Compliance

- **The EASA requirements and their AMCs (Acceptable Means of Compliance) constitute the framework of the EU Safety Management System**. This framework addresses the basic elements of ICAO defined in Appendix 2 of Annex 19, includes the elements of the compliance monitoring and promotes an integrated approach to the management of an organization.
- **This approach aims to encourage organizations to integrate information management safety and risk-based decision making in all its activities**, instead of overlaying another system on top of the management system and business structure.
- **The integration will allow to eliminate any duplication and take advantage of synergies through managing safety risks in multiple activities**. The organizations will be able, then. to determine the best way to structure their management systems to adapt them to your business and organizational needs.

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THANKS FOR YOUR ATTENTION